Calc III Section Notes with Answers

Spring 2026

Hui Sun
(hsun95@jh.edu)

April 6, 2026

Contents

1 The Geometry of Euclidean Spaces 3

2 Differentiation 12

3 Higher-Order Derivatives: Maxima and Minima 19

4 Vector-Valued Functions 24

5 Double and Triple Integrals 26

6 Change of Variables Formula 28

7 Integrals over Paths and Surfaces 33

Chapter 1

The Geometry of Euclidean Spaces

Week 1 (1/19-23)

Logistics

- TA: Hui.
- Email: hsun95@jh.edu.
- Drop-in Hours: Tuesday 2-3 PM, 5-6 pm, Krieger 211.
- Biweekly Quizzes: 10%.
- Attendance: 5%. (If you can't make it, email me).

Definition 1.1 (standard basis of $\mathbb{R}^3$)**.** The following vectors

$$i = (1, 0, 0), j = (0, 1, 0), k = (0, 0, 1)$$

are called the **standard basis** vectors of $\mathbb{R}^3$, and for any vector $a = (a_1, a_2, a_3) \in \mathbb{R}^3$, we can write

$$a = a_1 i + a_2 j + a_3 k$$

Definition 1.2 (dot product)**.** Let $v = (v_1, v_2, v_3), w = (w_1, w_2, w_3) \in \mathbb{R}^3$, the **dot product** $v \cdot w$ is given by

$$v \cdot w = v_1 w_1 + v_2 w_2 + v_3 w_3$$

Alternatively,

$$v \cdot w = \|v\|\|w\| \cos\theta$$

where

$$\theta = \arccos\left(\frac{v \cdot w}{\|v\|\|w\|}\right)$$

Definition 1.3 (length of vector)**.** Let $v = (v_1, v_2, v_3) \in \mathbb{R}^3$, the **length** or **norm** of v, denoted as $\|v\|$, is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{v\dot{v}}$$

Definition 1.4 (linear combination)**.** Let $v, w \in \mathbb{R}^3$, a **linear combination** of v, w is a sum

$$av + bw$$

for some $a, b \in \mathbb{R}$. One can generalize this definition to n vectors: let $v_1, v_2 \ldots, v_n \in \mathbb{R}^3$, a linear combination of these vectors is a finite sum

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for some $a_i \in \mathbb{R}, 1 \leq i \leq n$.

Proposition 1.1 (properties of the dot product)**.** Let $a, b, c \in \mathbb{R}^n$, then

(a) Nonnegativity: $a \cdot a \geq 0$, and $a \cdot a = 0$ if and only if $a = 0$.

(b) Scalar multiplication: let $\lambda \in \mathbb{R}$, then

$$\lambda(a \cdot b) = \lambda a \cdot b = a \cdot \lambda b$$

(c) Distributivity:

$$a \cdot (b + c) = a \cdot b + a \cdot c, \quad (a + b) \cdot c = a \cdot c + b \cdot c$$

(d) Symmetry: $a \cdot b = b \cdot a$.

Problem 1.1. Draw the following vectors in $\mathbb{R}^2$:

$$u = (1, 2), \quad v = (3, -2)$$

Compute $u + v, u - v$, and draw them in the plane.

Proof.

$$u + v = (4, 0), \quad u - v = (-2, 4)$$

□

Problem 1.2. Consider the following vectors in $\mathbb{R}^3$:

$$u = (1, 2, 3), \quad , v = (-2, 1, 4)$$

1. Compute their norms.
2. Two vectors $a, b \in \mathbb{R}^3$ are called **orthognal** if $a \cdot b = 0$. Are u, v orthogonal? If not, find a nonzero vector orthogonal to u.

Proof. 1.

$$\|u\| = (u \cdot u)^{\frac{1}{2}} = \sqrt{14}, \quad \|v\| = \sqrt{21}$$

2. We check

$$u \cdot v = -2 + 2 + 12 = 12 \neq 0$$

thus not orthogonal. A vector that is orthognal to u: $(-3, 0, 1)$. Note that this vector is **not** unique! For example, $(-1, -1, 1)$ is another such vector.

□

Problem 1.3. Can you express $w = (1, 2)$ as a linear combination of v_1, v_2 for difference choices of v_1, v_2?

1. $v_1 = (1, 1), v_2 = (-2, -2)$.
2. $v_1 = (2, 1), v_2 = (-1, 0)$.

Proof. 1. We first note that $(1, 1), (-2, -2)$ lie on the same line through the origin. Hence, any linear combination of v_1, v_2 will stay in this line, i.e., of the form (a, a), for some $a \in \mathbb{R}$. Therefore, it is impossible to write $w = (1, 2)$ as a linear combination of v_1, v_2.

2. Suppose $w = a_1 v_1 + a_2 v_2$ for some $a_1, a_2 \in \mathbb{R}$, then

$$\begin{cases} 2a_1 - a_2 = 1 \\ a_1 = 2 \end{cases} \Rightarrow \begin{cases} a_1 = 2 \\ a_2 = 3 \end{cases}$$

Thus we can write w as a linear combination of v_1, v_2:

$$w = 2v_1 + 3v_2$$

□

Problem 1.4. Let $u, v \in \mathbb{R}^3$, suppose that u, v are orthongal, show that

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Bonus: is the converse true? (meaning assuming $\|u + v\|^2 = \|u\|^2 + \|v\|^2$, is it true that $u \cdot v = 0$?)

Proof. We have

$$\begin{aligned} \|u + v\|^2 &= (u + v) \cdot (u + v) \\ &= u \cdot u + u \cdot v + v \cdot u + v \cdot v \\ &= \|u\|^2 + \|v\|^2 \end{aligned}$$

because $u \cdot v = v \cdot u = 0$. The converse is also true: we know by definition that

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2u \cdot v$$

given the assumption, we also have

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Thus equating them we get

$$\|u\|^2 + \|v\|^2 + 2u \cdot v = \|u\|^2 + \|v\|^2 \Rightarrow u \cdot v = 0$$

□

Week 2 (1/26-30)

Topics: determinant, cross product.

Definition 1.5 (determinant)**.** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2×2 matrix, the **determinant** of A is given by

$$\det(A) = ad - bc$$

Let $A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$ be a 3×3 matrix, the **determinant** of A is given by

$$\det(A) = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$

Definition 1.6 (cross product)**.** Let $a, b \in \mathbb{R}^3$, write $a = (a_1, a_2, a_3), b = (b_1, b_2, b_3)$, then the **cross product**

$$a \times b = \det \begin{bmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

where i, j, k are the standard vectors in $\mathbb{R}^3$.

Proposition 1.2 (properties of the cross product)**.** We have the following properties regarding the cross product: let $a, b \in \mathbb{R}^3$,

1. $a \times a = 0$.
2. $a \times b = -b \times a$.
3. $(a + b) \times c = a \times c + b \times c$, and $a \times (b + c) = a \times b + a \times c$.
4. $(\alpha a) \times b = \alpha(a \times b)$ for any $a \in \mathbb{R}$.
5. $a \times b$ is perpendicular to vectors a, b.
6. The length of the cross product is the area of the parallelogram spanned by a, b:

   $$\|a \times b\| = \|a\|\|b\| \sin\theta$$

   where $0 \leq \theta \leq \pi$ is the angle between them.
7. $a \times b = 0$ iff a, b are parallel or either a or b are 0.
8. The cross product is **not** associative! For example, compute

   $$(i \times i) \times j, \quad i \times (i \times j)$$

Proposition 1.3 (determinant and linear combination). Let $A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$ be a 3×3 matrix, let

$$a = (a_1, a_2, a_3), b = (b_1, b_2, b_3), c = (c_1, c_2, c_3)$$

If any of a, b, or c is a linear combination of the other two vectors, then $\det(A) = 0$. (Relevant topic: linear independence).

Problem 1.5. Let $\vec{u} = (1, 2, 3), \vec{v} = (0, 1, 1)$ be vectors in $\mathbb{R}^3$, compute the area of the parallelogram spanned by these two vectors.

Proof.

$$u \times v = \begin{bmatrix} i & j & k \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} = -i - j + k = (-1, -1, 1)$$

Thus the area of the parallelogram is

$$\|u \times v\| = \sqrt{3}$$

□

Problem 1.6. Compute the determinant of the following matrix A:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 3 & 1 & 1 \end{pmatrix}$$

Proof. Notice that the third row vector $(3, 1, 1)$ is the sum of the two row vectors above, hence by Proposition 1.3, we know we must have $\det(A) = 0$. □

Problem 1.7. Let $u = (1, 2, -1)$, find a nonzero vector that is orthogonal to u.

Proof. There are many choices, for example, $v = (1, 0, 1)$. □

Problem 1.8. Let $u = (1, 0), v = (2, -1)$,

1. Using the dot product, what is the angle between u, v? (You do not need to simplify your answer).
2. Can you find a nonzero vector in $\mathbb{R}^2$ that is orthogonal to both of u, v?
3. Can you find nonzero vectors that are orthogonal to any basis of $\mathbb{R}^2$? (Hint: take $u = (1, 0)$, can you draw the set of vectors orthogonal to u? A basis of $\mathbb{R}^2$ contains two noncolinear vectors).

Proof. 1. We have

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|} = \frac{2}{\sqrt{5}}$$

Hence

$$\theta = \arccos\left(\frac{2}{\sqrt{5}}\right)$$

2. Suppose there exists a nonzero vector $w = (w_1, w_2) \in \mathbb{R}^2$ such that $w \cdot u = w \cdot v = 0$, then this implies

$$w \cdot u = w_1 = 0, \quad w \cdot v = 2w_1 - w_2 = 0 \Rightarrow w_1 = w_2 = 0$$

Thus the only vector orthogonal to u, v is the zero vector, hence the answer is no.

3. Let $\{a, b\}$ be a basis of $\mathbb{R}^2$, where $a = (a_1, a_2), b = (b_1, b_2)$, we know they are noncolinear, hence

$$\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$$

Then suppose w is orthogonal to both a, b. Then (one can draw a picture), w should be of the form

$$w = (\lambda a_2, -\lambda a_1) = (\zeta b_2, -\zeta b_1)$$

for some $\lambda, \zeta \in \mathbb{R}$. But this implies

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{\zeta}{\lambda}$$

which is a contradiction to the above. This concludes the proof that no nonzero vector can be orthogonal to a basis in $\mathbb{R}^2$.

□

Problem 1.9. Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Compute the determinant of A.

Proof. We have $\det(A) = 10$. □

Problem 1.10. Construct a nonzero 3×3 matrix A such that $\det(A) = 0$.

Proof.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$$

□

Week 3 (2/2-2/6)

Topics: parametric equations, multivariable functions, and level sets.

Definition 1.7 (matrix addition, multiplication)**.** Let A, B be $m \times n$ matrices, C be $n \times k$ matrix as follows

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & b_{mn} \end{pmatrix}$$

$$C = \begin{pmatrix} c_{11} & \dots & c_{1k} \\ \vdots & \ddots & \vdots \\ c_{n1} & \dots & c_{nk} \end{pmatrix}$$

then **matrix addition** $A + B$ defined as

$$A + B := \begin{pmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \dots & a_{mn} + b_{mn} \end{pmatrix}$$

and **matrix multiplication** is defined as

$$AC = \begin{pmatrix} \sum_{j=1}^{n} a_{1j} c_{j1} & \dots & \sum_{j=1}^{n} a_{1j} c_{jk} \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^{n} a_{mj} c_{j1} & \dots & \sum_{j=1}^{n} a_{mj} c_{jk} \end{pmatrix}$$

Remark. *Given matrices A, B, for the matrix multiplication AB to be well-defined,*

$$\textit{number of columns of } A = \textit{number of rows of } B$$

Definition 1.8 (matrix as linear transformation)**.** Let A be an $m \times n$ matrix

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

and let $x \in \mathbb{R}^n$, where $x = (x_1, \dots, x_n)$. Then A applied to x as a linear transformation is given by

$$Ax = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{n} a_{1j} x_j \\ \vdots \\ \sum_{j=1}^{n} a_{mj} x_j \end{pmatrix}$$

where

$$\sum_{j=1}^{n} a_{1j} x_j = a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n, \quad \dots$$

Definition 1.9 (image, graph). The **image** of a function $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$ is a subset of $\mathbb{R}^m$,

$$\text{Image}(f) = \{f(x) \in \mathbb{R}^m : x \in U\}$$

and the **graph** of f is a subset of $\mathbb{R}^{n+m}$,

$$\text{Graph}(f) = \{(x, f(x)) : x \in U\}$$

Definition 1.10 (level set). Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$, and $c \in \mathbb{R}$ be some constant. Then the **level set** of f at c is the set $\mathcal{L}_c$

$$\mathcal{L}_c := \{x \in U : f(x) = c\} \subset \mathbb{R}^n$$

Definition 1.11 (Equation of a line). A **line** l in $\mathbb{R}^3$ through the tip of $a = (a_1, a_2, a_3)$ pointing in the direction of a vector $v = (v_1, v_2, v_3)$ is given by

$$l(t) = a + tv = (a_1 + tv_1, a_2 + tv_2, a_3 + tv_3)$$

where $t \in \mathbb{R}$. Alternatively, a line passing through two points $P = (x_1, y_1, z_1), Q = (x_2, y_2, z_2)$ is given by

$$l(t) = (x(t), y(t), z(t))$$

where

$$\begin{cases} x(t) = x_1 + (x_2 - x_1)t \\ y(t) = y_1 + (y_2 - y_1)t \\ z(t) = z_1 + (z_2 - z_1)t \end{cases}$$

Definition 1.12 (Plane in $\mathbb{R}^3$). If a plane P passes through some point (x_0, y_0, z_0), and $n = (A, B, C)$ is a vector orthogonal to the plane, then the plane P is given by the equation:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Remark. *A point P in the plane and a normal vector to the plane uniquely determine a plane in $\mathbb{R}^3$. Equivalently, three noncolinear points uniquely determine a plane.*

Problem 1.11. Compute the plane containing all three points:

$$(1, 0, 2), \quad (2, -1, 0), \quad (-1, 2, 3)$$

Proof. Let $A = (1, 0, 2), B = (2, -1, 0), C = (-1, 2, 3)$, then consider two vectors in this plane

$$AB = (1, -1, -2), AC = (-2, 2, 1)$$

Then taking their cross product we find a normal vector to this plane:

$$AB \times AC = \begin{bmatrix} i & j & k \\ 1 & -1 & -2 \\ -2 & 2 & 1 \end{bmatrix} = 3i + 3j + 0k = (3, 3, 0)$$

Thus using the definition above, and point A, we know the formula is given by

$$3(x - 1) + 3(y) = 0$$

One can simplify this to

$$x + y - 1 = 0$$

□

Problem 1.12. (a) Find the equation of the line through $(1, 1, 0)$ in the direction of $2\mathbf{i} - \mathbf{k}$.

(b) Find the equation of the line passing through $(0, 1, 1)$ and $(0, 1, 0)$.

(c) Find an equation for the plane perpendicular to the vector $(-1, 1, -1)$ and passing through the point $(1, 1, 1)$.

Proof. (a) The equation is given by

$$l_1(t) = (1, 1, 0) + t(2, 0, -1), \quad t \in \mathbb{R}$$

(b) The equation is given by

$$l_2(t) = (0, 1, 1) + t(0, 0, -1), \quad t \in \mathbb{R}$$

(c) The equation of the plane is given by

$$(x - 1, y - 1, z - 1) \cdot (-1, 1, -1) = 0$$

simplifying we get

$$x - y + z = 1$$

□

Chapter 2

Differentiation

Week 4 (2/9-2/13)

Topic: limits.

Definition 2.1 (limit). Let $f : A \subset \mathbb{R}^n \to \mathbb{R}^m$, where A is open, let x_0 be in A or be a boundary point of A and N be a neighborhood of a point $b \in \mathbb{R}^m$. Now let x approach x_0, f is said to be **eventually in** N if there exists a neighborhood U of x_0 such that

$$\text{if } x \in U, \text{ then } f(x) \in N$$

If f is eventually in N for *any* neighborhood N around b, then the **limit** of f as $x \to x_0$ exists, denoted as

$$\lim_{x \to x_0} f(x) = b$$

Alternatively, if the limit exists, then $\lim_{x \to x'} f(x) = b$ is when $x = (x_1, x_2, \ldots, x_n) \to x' = (x'_1, x'_2, \ldots, x'_n)$ from **all directions**, and $f(x)$ approaches $b = (b_1, \ldots, b_m)$.

Definition 2.2 (continuity). Let $f : A \subset \mathbb{R}^n \to \mathbb{R}^m$ is said to be **continuous** at $x_0 \in A$ if

$$\lim_{x \to x_0} f(x) = f(x_0)$$

And f is called continuous if f is continuous at every $x_0 \in A$.

Example 2.1. The limit doesn't need to exist! For example, let

$$H(x) = \begin{cases} 1, x \geq 0 \\ -1, x < 0 \end{cases}$$

Note the limit doesn't exist at $x = 0$.

Problem 2.1. For the following functions, find their (1) image, (2) graph, (3) **draw their level sets**.

1. Let $f : \mathbb{R} \to \mathbb{R}$, and $f(x) = x^2 + 1$.
2. Let $g : \mathbb{R}^2 \to \mathbb{R}$, and $g(x, y) = x^2 + y^2$.

Proof. 1. Image$(f) = \{x^2 + 1 : x \in \mathbb{R}\}$, and Graph$(f) = \{(x, x^2 + 1) : x \in \mathbb{R}\}$.

2. Image$(g) = \{x^2 + y^2, (x, y) \in \mathbb{R}^2\}$, and Graph$(g) = \{(x, y, x^2 + y^2) : (x, y) \in \mathbb{R}^2\}$.

□

Problem 2.2. Compute the following limits:

1. $$\lim_{(x,y)\to(0,0)} \frac{\sin xy}{y}$$

   (Hint: try writing $\frac{\sin xy}{y} = \frac{\sin xy}{xy} \cdot x$, and recall $\lim_{t\to 0} \frac{\sin t}{t} = 1$).

2. $$\lim_{(x,y)\to(0,0)} \frac{e^{xy} - 1}{y}$$

3. $$\lim_{(x,y)\to(0,0)} \frac{(x-y)^2}{x^2 + y^2}$$

Proof. 1. Following the hint, we see

$$\lim_{(x,y)\to(0,0)} \frac{\sin xy}{y} = \lim_{(x,y)\to(0,0)} \frac{\sin xy}{xy} x = \lim_{x\to 0} x = 0$$

2. This one uses the exact same trick:

$$\lim_{(x,y)\to(0,0)} \frac{e^{xy} - 1}{xy} \cdot y = 0$$

3. First letting $x \to 0$ along $y = 0$, we see the limit is 1; letting $x = y \to 0$, we see the limit is 0, thus the limit doesn't exist!

□

Problem 2.3. Compute the limit of the following functions:

1. $$\lim_{(x,y)\to(0,0)} \frac{x}{x+y}$$

2. $$\lim_{(x,y)\to(0,0)} \frac{xy}{x+y}$$

   (Hint: try considering $y = x^2 - x$ and $y = x$)

3. $$\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{x+y}$$

Proof. 1. First fix $x = 0$, let $y \to 0$, then the limit is 0; now fix $y = 0$, let $x \to 0$, the limit is 1. The limit doesn't exist!

2. Consider $y = x^2 - x$, (as $x \to 0, y \to 0$), then

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x+y} = \lim_{x\to 0} \frac{x^3 - x^2}{x^2} = \lim_{x\to 0} x - 1 = -1$$

and consider $y = x$, we see the limit is 0, thus the limit doesn't exist!

Warning 2.1. 2 does not follow from 1! A student suggests a proof: $\lim_{(x,y)\to(0,0)} = \frac{x}{x+y} \cdot y$, and by 1, the limit $\frac{x}{x+y}$ doesn't exist, this implies the limit of $\frac{xy}{x+y}$ also doesn't exist. This argument is not correct! Consider the following counterexample: $\lim_{y\to 0} \frac{1}{y}$ doesn't exist, but the limit

$$\lim_{y\to 0} \frac{1}{y} \cdot y = 1$$

exists! More concretely, if you multiply by any function that doesn't tend to 0, the argument follows, but it doesn't work when the function tends to 0!

3. We see that

$$\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{xy} \frac{xy}{x+y}$$

Note that the limit of $\sin(xy)/(xy) = 1$, but the second one doesn't exist, thus the limit doesn't exist! $\square$

Idea 2.2. How to find a a limit $\lim_{x\to x_0} f(x)$:

- Step 1: Guess what the limit should be.
- Step 2: Try from approaching x_0 from different directions.
- Step 3: Try to replace terms with expressions you are familiar with.

Week 5 (2/16-2/20)

Topic: derivatives.

Definition 2.3 (partial derivative)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, where U is open. Then the **partial derivative** with respect to x_i is defined by

$$\frac{\partial f}{\partial x_i}(x_1, \ldots, x_n) = \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h}$$

where $e_i = (0, \ldots, 1, \ldots, 0)$ with 1 in the ith coordinate.

Definition 2.4 (differentiability in two variables)**.** Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$, then f is **differentiable** at $(x_0, y_0) \in U$ if

(1) $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist at (x_0, y_0)

(2)
$$\lim_{(x,y) \to (x_0, y_0)} \frac{f(x,y) - f(x_0,y_0) - \left[\frac{\partial f}{\partial x}(x_0,y_0)\right](x - x_0) - \left[\frac{\partial f}{\partial y}(x_0,y_0)\right](y - y_0)}{\|(x,y) - (x_0,y_0)\|} = 0$$

If f is differentiable at (x_0, y_0), then the derivative of f at (x_0, y_0) is the 1×2 matrix

$$\begin{bmatrix} \frac{\partial f}{\partial x}(x_0,y_0) & \frac{\partial f}{\partial y}(x_0,y_0) \end{bmatrix}$$

Warning 2.3. It is not enough only require the partial derivatives exist at the point. Consider the following function:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

The partitial derivatives exist at $(0,0)$, but f is not differentiable.

Definition 2.5 (tangent plane)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be differentiable at (x_0, y_0), then the **tangent plane** to the graph f at (x_0, y_0) is the given by

$$z = f(x_0,y_0) + \frac{\partial f}{\partial x}(x_0,y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0,y_0)(y - y_0)$$

Remark. *The plane given above intersects the graph of f at the point $(x_0, y_0, f(x_0, y_0))$.*

Definition 2.6 (differentiability in the general setting)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$, then f is differentiable at $x_0 \in U$ if

(1) the partial derivatives $\frac{\partial f_i}{\partial x_j}$ exist for all $1 \leq i \leq m, 1 \leq j \leq n$.

(2)
$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - T(x - x_0)\|}{\|x - x_0\|} = 0$$

where $T = Df(x_0)$ is the $m \times n$ matrix

$$Df(x_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

Definition 2.7 (gradient)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, the **gradient** $\nabla f(x)$ is a special case of the general case above when $m = 1$, i.e., it is a $1 \times n$ matrix

$$Df(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Proposition 2.1. Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x_0, then the derivative of f at x_0 is an $m \times n$ matrix $Df(x_0) = \left(\frac{\partial f_i}{\partial x_j}\right)_{ij}$. The derivative follows the same properties as derivative for single variable functions:

1. Let $c \in \mathbb{R}$, then
$$D(cf)(x_0) = cDf(x_0) \qquad \text{(multiplication of a matrix by constant } c\text{)}$$

2. Let $g : U \subset \mathbb{R}^n \to \mathbb{R}^m$ also be differentiable at x_0, then
$$D(f + g)(x_0) = Df(x_0) + Dg(x_0) \qquad \text{(sum of two matrices)}$$

3. Let $h_1 : U \subset \mathbb{R}^n \to \mathbb{R}, h_2 : U \subset \mathbb{R}^n \to \mathbb{R}$,then
$$D(h_1 h_2)(x_0) = Dh_1(x_0)h_2(x_0) + h_1(x_0)Dh_2(x_0) \qquad \text{(product rule)}$$
and if $h_2 \neq 0$ on U.
$$D(h_1/h_2)(x_0) = \frac{Dh_1(x_0)h_2(x_0) - h_1(x_0)Dh_2(x_0)}{h_2^2(x_0)} \qquad \text{(quotient rule)}$$

4. Let $g : U \subset \mathbb{R}^n \to \mathbb{R}^m, f : V \subset \mathbb{R}^m \to \mathbb{R}^p$ such that $g(U) \subset V$, then
$$D(f \circ g)(x_0) = Df(g(x_0))Dg(x_0) \qquad \text{(chain rule)}$$

Proposition 2.2 (special case of chain rule)**.** Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a function in 3 variables u, v, w, namely, $z = f(u, v, w)$, and if $u = g_1(x), v = g_2(x), w = g_3(x)$, then

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial z}{\partial w}\frac{\partial w}{\partial x}$$

Problem 2.4. Compute the plane tangent to the graph of $f(x, y) = x^2y + 2xy - y^2$ at $(1, 2, 2)$.

Proof. We have

$$\frac{\partial f}{\partial x}(1,2) = 2xy + 2y|_{(1,2)} = 8, \quad \frac{\partial f}{\partial y}(1,2) = x^2 + 2x - 2y|_{(1,2)} = -1$$

and $f(1, 2) = 2$, thus the plane is given by

$$z = 2 + 8(x - 1) - (y - 2)$$

i.e., $z = 8x - y - 4$. □

Problem 2.5. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by

$$f(x, y, z) = x^2y + y\sin(z) + ze^x.$$

Compute the gradient of f at $(1, 2, 0)$.

Proof. You can compute the partial derivatives

$$\begin{aligned}\frac{\partial f}{\partial x} &= 2xy + ze^x,\\ \frac{\partial f}{\partial y} &= x^2 + \sin(z),\\ \frac{\partial f}{\partial z} &= y\cos(z) + e^x.\end{aligned}$$

thus the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (2xy + ze^x, x^2 + \sin z, y\cos z + e^x)$$

Hence

$$\nabla f(1, 2, 0) = (4, 1, 2 + e)$$

□

Problem 2.6. Given $w = 2xy + z$, where $x = 2t, y = -t, z = t^2$, compute $\frac{\partial w}{\partial t}$.

Proof. Using chain rule, we write

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial y}{\partial t}\frac{\partial y}{\partial t} + \frac{\partial z}{\partial t}\frac{\partial z}{\partial t} = 4y - 2x + 2t = -6t$$

□

Problem 2.7 (2.5, Q7). Let $f(u, v) = (\tan(u - 1) - e^v, u^2 - v^2)$ and

$$g(x, y) = (e^{x-y}, x - y).$$

Calculate $f \circ g$ and

$$D(f \circ g)(1, 1).$$

Proof. We have

$$f \circ g(x,y) = (\tan(e^{x-y}-1) - e^{x-y}, e^{2(x-y)} - (x-y)^2)$$

and $g(1,1) = (1,0)$, thus using chain rule, we have

$$D(f \circ g)(1,1) = Df(1,0)Dg(1,1)$$

where

$$Df(u,v) = \begin{bmatrix} \sec^2(u-1) & -e^v \\ 2u & -2v \end{bmatrix}, \quad Dg(x,y) = \begin{bmatrix} e^{x-y} & -e^{x-y} \\ 1 & -1 \end{bmatrix}$$

Hence

$$\begin{aligned} D(f \circ g)(1,1) &= \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

□

Problem 2.8 (2.5, Q8). Let $f(u,v,w) = (e^{u-w}, \cos(v+u) + \sin(u+v+w))$ and $g(x,y) = (e^x, \cos(y-x), e^{-y})$. Calculate $f \circ g$ and $D(f \circ g)(0,0)$.

Proof. We have

$$f \circ g = (e^{e^x - \cos(y-x)}, \cos(e^x + \cos(y-x)), \sin(e^x + e^{-y} + \cos(y-x)))$$

and $g(0,0) = (1,1,1)$. Thus

$$D(f \circ g)(0,0) = Df(1,1,1)Dg(0,0)$$

where

$$Df(u,v,w) = \begin{bmatrix} e^{u-w} & 0 & -e^{u-w} \\ -\sin(v+u) + \cos(u+v+w) & -\sin(v+u) + \cos(u+v+w) & \cos(u+v+w) \end{bmatrix}$$

and

$$Dg(x,y) = \begin{bmatrix} e^x & 0 \\ \sin(y-x) & -\sin(y-x) \\ 0 & -e^{-y} \end{bmatrix}$$

Thus

$$\begin{aligned} D(f \circ g)(0,0) &= \begin{bmatrix} 1 & 0 & -1 \\ -\sin 2 + \cos 3 & -\sin 2 + \cos 3 & \cos 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ -\sin 2 + \cos 3 & -\cos 3 \end{bmatrix} \end{aligned}$$

□

Chapter 3

Higher-Order Derivatives: Maxima and Minima

Week 7 (3/2-3/6)

Definition 3.1 (quadratic form). A function $g : \mathbb{R}^n \to \mathbb{R}$ is called a **quadratic form** if it is given by

$$g(h_1, \ldots, h_n) = \sum_{i,j=1}^{n} a_{ij} h_i h_j$$

where (a_{ij}) is an $n \times n$ matrix. We can also write g as follows:

$$g(h_1, \ldots, h_n) = [h_1, \ldots, h_n] \begin{bmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n_1} & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$

Definition 3.2 (Hessian matrix). Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, and suppose all the second-order partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ exist, then the Hessian matrix of f is the $n \times n$ matrix given by

$$Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

The Hessian as a quadratic function is defined by

$$Hf(x)(h) = \frac{1}{2} \begin{bmatrix} h_1 & \ldots & h_n \end{bmatrix} Hf(x) \begin{bmatrix} h_1 \\ \ldots \\ h_n \end{bmatrix}$$

where $h = (h_1, \ldots, h_n)$.

Definition 3.3 (degenerate/nondegenerate points). Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$ be of C^2, let (x_0, y_0) be a critical point. We define the **discriminant**, $\mathcal{D}$,of the Hessian by

$$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

If $\mathcal{D} \neq 0$, the critical point (x_0, y_0) is called **nondegenerate**; if $\mathcal{D} = 0$, the point (x_0, y_0) is called **degenerate**.

Definition 3.4 (positive, negative-definite). A quadratic form $g : \mathbb{R}^n \to \mathbb{R}$ is called **positive-definite** if $g(h) \geq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$. Similarly, g is **negative-definite** if $g(h) \leq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$. (The matrix is positive-definite iff it is symmetric $A^T = A$ and the eigenvalues are nonnegative).

Definition 3.5 (bounded set). A set $A \subset \mathbb{R}^n$ is said to be **bounded** if there is a number $M > 0$ such that $\|x\| \leq M$ for all $x \in A$.

Proposition 3.1 (extremums are critical points). Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable, where U is open. If x_0 is a local extremum, then $Df(x_0) = 0$.

Proposition 3.2 (extremum). Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ be in C^3, and x_0 is a critical point of f. If the Hessian $Hf(x_0)$ is positive-definite, then x_0 is a local minimum of f; if $Hf(x_0)$ is negative-definite, then x_0 is a local maximum.

Proposition 3.3 (local minimum). Let $f(x, y)$ be of C^2, and U is open in $\mathbb{R}^2$. A point (x_0, y_0) is a strict local **minimum** of f if the following conditions hold:

1. $$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

2. $$\mathcal{D}(x_0, y_0) > 0$$

   where $\mathcal{D}$ is the **discriminant** of the Hessian, defined by

   $$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

   where Hf is the 2×2 Hessian matrix.

3. $$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$$

If $\frac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0$ in 3, then it becomes a local **maximum**.

Proposition 3.4 (saddle points). Let $f(x, y) : U \subset \mathbb{R}^2 \to \mathbb{R}$ be of C^2, if $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$, and $\mathcal{D}(x_0, y_0) < 0$, where $\mathcal{D}$ is the discriminant, then the critical point (x_0, y_0) is a saddle point, i.e., neither a maximum or a minimum.

Proposition 3.5 (continuous functions attain extremum on closed bounded sets)**.** Let $f : D \to \mathbb{R}$ be continuous, where D is closed and bounded in $\mathbb{R}^n$. Then f assumes its absolute maximum and absolute minimum values at some point $x_0, x_1 \in D$.

Problem 3.1. Is the following matrix positive-definite?

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Proof. It is not! Consider the vector $(0, 1)$, we have

$$[0, 1] \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [0, 1] \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1$$

□

Problem 3.2. Find the critical point of $f(x, y) = y + x \sin y$ and classify whether it is a local max/min or a saddle point.

Proof. We compute

$$f_x = \sin y, \quad f_y = 1 + x \cos y$$

Setting them both to 0 gives

$$(1, n\pi), \text{ when } n \text{ is odd}, \quad (-1, n\pi) \text{ when } n \text{ is even}$$

Now we compute the discriminant:

$$f_{xx} = 0, \quad f_{xy} = f_{yx} = \cos y, \quad f_{yy} = -x \sin y$$

Thus

$$\mathcal{D} = \det \begin{bmatrix} 0 & \cos y \\ \cos y & -x \sin y \end{bmatrix} = -\cos^2 y$$

Thus $\mathcal{D} < 0$ for all the critical points, hence they are saddle points! □

Week 8 (3/9-3/13)

Proposition 3.6. If f has a maximum or minimum at x_0 when constrained to a surface S, then $\nabla f(x_0)$ is perpendicular to S at x_0.

Proposition 3.7. Let $f, g : U \subset \mathbb{R}^n \to \mathbb{R}$ and $\vec{x_0} \in U$ such that $g(x_0) = c$. Let $\mathcal{L}_c$ be the level set for g with value c and assume

$$\nabla g(\vec{x_0}) \neq \vec{0}.$$

If f restricted to $\mathcal{L}_c$ has a local minimum or maximum on $\mathcal{L}_c$ at x_0, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0).$$

Problem 3.3 (Marsden-Tromba, III. 2)**.** Let $f(x, y, z) = x - y + z$, find the extremum of f subject to the constraint $x^2 + y^2 + z^2 = 2$.

Proof. We compute the gradient of f and $g(x, y, z) = x^2 + y^2 + z^2$:

$$\nabla f(x, y, z) = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \quad \nabla g(x, y, z) = \begin{bmatrix} 2x & 2y & 2z \end{bmatrix}$$

By the proposition above, we need to have

$$\begin{aligned} \lambda &= 2x \\ -\lambda &= 2y \\ \lambda &= 2z \end{aligned}$$

Equating all λ gives

$$x = z = -y$$

Plugging in the constraint we get

$$3x^2 = 2$$

Hence two critical points

$$\left(\sqrt{\frac{2}{3}}, -\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right), \quad \left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}, -\sqrt{\frac{2}{3}}\right)$$

Plugging them in we find the extremums are $\sqrt{6}$ and $-\sqrt{6}$. □

Problem 3.4 (Marsden-Tromba, III. 16)**.** Use Lagrange multipliers to find the distance from the point $(2, 0, -1)$ to the plane $3x - 2y + 8z + 1 = 0$.

Proof. Let (x, y, z) be a point on the plane, then we would like the minimize the function

$$f(x, y, z) = (x - 2)^2 + y^2 + (z + 1)^2$$

with the constraint

$$3x - 2y + 8z + 1 = 0$$

Writing $g(x, y, z) = 3x - 2y + 8z + 1$, we do the exact same thing as we did above:

$$\nabla f = \begin{bmatrix} 2x - 4 & 2y & 2z + 2 \end{bmatrix}, \quad \nabla g = \begin{bmatrix} 3 & -2 & 8 \end{bmatrix}$$

Then setting

$$\begin{aligned} 2x - 4 &= 3\lambda \\ 2y &= -2\lambda \\ 2z + 2 &= 8\lambda \end{aligned}$$

One can then replace x, y, z with expressions in λ in the constraint:

$$3x - 2y + 8z + 1 = 0$$

which gives

$$\lambda = 2/77$$

Then plugging in λ to solve for x, y, z, the $\sqrt{f(x, y, z)}$ is the final answer. (I am too lazy to do the computation). □

Problem 3.5. Find the absolute maximum of $f(x, y) = 2x^2y$ inside the disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Proof. We first consider the points on the interior $\{(x, y) : x^2 + y^2 < 1\}$. We do so by finding the critical points $\nabla f(x, y) = 0$.

$$\nabla f(x, y) = \begin{bmatrix} 4xy & 2x^2 \end{bmatrix} = 0$$

which implies $(x, y) = (0, 0)$. And $f(0, 0) = 0$.

Now we consider the boundary. Let $g(x, y) = x^2 + y^2$, using Lagrange, we set up

$$\nabla f = \lambda \nabla g \Rightarrow \begin{bmatrix} 4xy & 2x^2 \end{bmatrix} = \lambda \begin{bmatrix} 2x & 2y \end{bmatrix}$$

This implies

$$\begin{cases} 4xy = 2\lambda x \\ 2x^2 = 2\lambda y \end{cases} \Rightarrow \begin{cases} 2y = \lambda \\ x^2 = \lambda y \end{cases}, \quad x^2 + y^2 = 1$$

This implies

$$(x, y) = \left(\frac{1}{\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}} \right)$$

thus

$$f(x, y) = \frac{2\sqrt{6}}{9}$$

comparing it with $f(0, 0) = 0$, we see

$$\left(\frac{1}{\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}} \right) \text{ is an absolute maximum.}$$

□

Chapter 4

Vector-Valued Functions

(Spring Break)

Topics: (1) Acceleration and Arc Length, (2) Vector Fields.

Proposition 4.1 (Newton's second law)**.** Let $c(t)$ be a path of a particle with mass m and $a(t) = c''(t)$ be the acceleration, then

$$F(c(t)) = ma(t)$$

where F is the force applying on the particle.

Definition 4.1 (arc length)**.** Let $c(t) = (x(t), y(t), z(t))$ be a path, then the length of the path in $\mathbb{R}^3$ from $t_0 \leq t \leq t_1$ is

$$\begin{aligned} L_{t_0 \to t_1}(c) &= \int_{t_0}^{t_1} \left(x'(t)^2 + y'(t)^2 + z'(t)^2\right)^{\frac{1}{2}} dt \\ &= \int_{t_0}^{t_1} \|c'(t)\| dt \end{aligned}$$

More generally, if $c(t) = (x_1(t), \ldots, x_n(t))$ is a path in $\mathbb{R}^n$, then

$$L_{t_0 \to t_1}(c) = \int_{t_0}^{t_1} \left(\sum_{i=1}^{n} x_i'(t)^2\right)^{\frac{1}{2}} dt$$

Definition 4.2 (vector field)**.** A vector field is a function $F : A \subset \mathbb{R}^n \to \mathbb{R}^n$ that assigns $x \in \mathbb{R}^n$ to another vector $F(x) \in \mathbb{R}^n$.

Problem 4.1. Find the velocity, speed, and acceleration of the following path at $t = 0$:

$$c(t) = (\cos t, 2t, -\sin t)$$

Proof. The velocity is

$$c'(t) = (-\sin t, 2, -\cos t), \quad c'(0) = (0, 2, -1)$$

And the speed is

$$\|c'(t)\| = \sqrt{5}$$

□

Problem 4.2. Find the length of the curve above from $t = 0$ to $t = 2$.

Proof.

$$\begin{aligned} L_{0\to 2}\|c'(t)\|dt &= \int_0^2 \left(\sin^2 t + 4 + \cos^2 t\right)^{\frac{1}{2}} dt \\ &= \int_0^2 \sqrt{5}dt \\ &= 2\sqrt{5} \end{aligned}$$

□

Chapter 5

Double and Triple Integrals

Week 10 (3/23-3/27)

Topics: (1) Divergence and Curl, (2) Double integrals.

Definition 5.1 (flow line). Let F be a vector field, a flow line of F is a path $c(t)$ satisfying

$$c'(t) = F(c(t))$$

(Tangent vector of the path coincides with the given vector field F).

Definition 5.2 (divergence). Let F be a vector field in $\mathbb{R}^3$ $F = (F_1, F_2, F_3)$, the divergence of F is the **scalar field** (assigns one number to an given point (x, y, z)),

$$\operatorname{div} F := \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

More generally, if $F = (F_1, \ldots, F_n)$ is a vector field on $\mathbb{R}^n$, its divergence is

$$\operatorname{div} F = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n}$$

Remark. *We write the divergence as $\nabla \cdot F$ because*

$$\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n} \right)$$

and if $F = (F_1, \ldots, F_n)$,

$$\operatorname{div} F = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n} = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n} \right) \cdot (F_1, \ldots, F_n) = \nabla \cdot F$$

Definition 5.3 (curl). Let F be a vector field in $\mathbb{R}^3$, writing $F = (F_1, F_2, F_3)$, the **curl** of F is the vector field

$$\operatorname{curl} F := \nabla \times F = \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix}$$

If $\operatorname{curl} F = 0$, then we say the vector field is **irrotational**.

Warning 5.1. Irrotational does not mean the vector field doesn't look like rotations.

Proposition 5.1 (gradient is irrotational)**.** Let $f \in C^2$, viewing ∇f as a vector field, then

$$\nabla \times (\nabla f) = 0$$

Proposition 5.2 (divergence of a curl vanishes)**.** For any C^2 vector field F,

$$\nabla \cdot (\nabla \times F) = 0$$

Problem 5.1. Show that the vector field $V(x, y, z) = (x^2, -y, z)$ is not the curl of any vector field F. In other words, there is no vector field F such that

$$V = \operatorname{curl} F$$

Proof. By the above proposition, if there exists F such that

$$V = \operatorname{curl} F$$

then

$$\nabla \cdot V = 0$$

However,

$$\nabla \cdot V(x, y, z) = 2x \neq 0 \text{ if } x \neq 0$$

this is a contradiction, hence no such F exists. □

Problem 5.2 (Marsden-Tromba, IV.4, 21)**.** Let $F(x, y, z) = (x^2, x^2y, z + zx)$. Can there exist a function $f : \mathbb{R}^3 \to \mathbb{R}$ such that $F = \nabla f$?

Proof. No. Suppose there exists such f, then the gradient must be irrotational. In other words, we must have

$$\nabla \times F = 0$$

Thus we compute the curl of F:

$$\nabla \times F = (-2yz, z, 0)$$

and it is not identically 0. □

Chapter 6

Change of Variables Formula

Week 11 (3/30-4/3)

Topics: (1) Double and Triple Integration, (2) Change of Variables Theorem.

Definition 6.1 (injective, surjective)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a map, then T is **injective, or one-to-one,** if for $x, y \in \mathbb{R}^2$

$$Tx = Ty$$

then

$$x = y.$$

T is called **surjective or onto** if for all $y \in \mathbb{R}^2$, there exists $x \in \mathbb{R}^2$ such that

$$Tx = y$$

If T is both injective and bijective, then we say T is **bijective**.

Definition 6.2 (linear map)**.** Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$, then T is linear if and only if for all $x, y \in \mathbb{R}^2, \lambda \in \mathbb{R}$.

$$\begin{cases} T(x+y) = Tx + Ty \\ T(\lambda x) = \lambda Tx \end{cases}$$

Definition 6.3 (Jacobian Determinant)**.** Let $T : D^* \subset \mathbb{R}^2 \to \mathbb{R}^2$ be of C^1 defined by

$$T : \begin{pmatrix} u \\ v \end{pmatrix} \mapsto \begin{pmatrix} x(u,v) \\ y(u,v) \end{pmatrix}$$

The **Jacobian determinant** of T, denoted as $\frac{\partial(x,y)}{\partial(u,v)}$ is the determinant of the matrix $DT(u,v)$:

$$\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

Proposition 6.1 (Fubini's Theorem for rectangles)**.** Let f be a continuous function on a rectangular domain $R = [a,b] \times [c,d]$, then

$$\int_a^b \int_c^d f(x,y)dydx = \int_c^d \int_a^b f(x,y)dxdy = \int_R f(x,y)dA$$

Proposition 6.2 (Fubini's Theorem for general regions). Suppose D is a set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, and similarly for $x \in [a, b]$, $\varphi_1(x) \leq y \leq \varphi_2(x)$. If f is continuous on D, then

$$\iint_D f(x,y)dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)dydx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x,y)dxdy$$

Proposition 6.3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map, then there exists a 2×2 matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ such that

$$Tx = Ax$$

Proposition 6.4. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map, then T is injective if and only if it is also surjective if and only if $\det(A) \neq 0$, where A is the matrix associated to T.

Theorem 6.1 (change of variables formula). Let D, D^* be elementary regions in $\mathbb{R}^2$, suppose $T : D^* \to D$ is bijective. Then for any integral function $f : D \to \mathbb{R}$, the **change of variable formula** states

$$\iint_D f(x,y)dxdy = \iint_{D^*} f(x(u,v), y(u,v)) \left|\det(J)\right| dudv$$

where

$$\det(J) = \left|\frac{\partial(x,y)}{\partial(u,v)}\right|$$

is the determinant of the Jacobian matrix.

Corollary 6.1. An immediate corollary of the above theorem is the change of variables into polar coordinates:

$$\iint_D f(x,y)dxdy = \iint_{D^*} f(r\cos\theta, r\sin\theta)rdrd\theta$$

Corollary 6.2. Let W, W^* be elementary regions in $\mathbb{R}^3$, and suppose $T : W^* \to W$ is bijective. Then the change of varialbes formula for triple integrals states:

$$\iiint_W f(x,y,z)dxdydz = \iiint_{W^*} f(x(u,v,w), y(u,v,w), z(u,v,w)) \mid \det(J) \mid dudvdw$$

where

$$\det(J) = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}$$

Problem 6.1. Change the order of integration to $dydx$ for the following function:

$$\int_0^1 \int_{e^y}^e \frac{x}{\ln x} dxdy$$

Proof. It should be

$$\int_1^e \int_0^{\ln x} \frac{x}{\ln x} dydx$$

□

Problem 6.2. Change the order of integrations for the following functions:

1. $$\int_0^1 \int_x^1 f(x,y)dydx$$
2. $$\int_0^1 \int_{-y}^{y^3} f(x,y)dxdy$$
3. $$\int_0^3 \int_{2x}^6 f(x,y)dydx$$
4. $$\int_0^1 \int_{-\sqrt{y}}^{y^2} f(x,y)dxdy$$
5. $$\int_0^8 \int_{\sqrt[3]{y}}^2 f(x,y)dxdy$$
6. $$\int_0^1 \int_{\ln y}^1 f(x,y)dxdy$$

Proof.

1. It should be
$$\int_0^1 \int_0^y f(x,y)dxdy$$
2. It should be
$$\int_{-1}^0 \int_{-x}^1 f(x,y)dydx + \int_0^1 \int_{\sqrt{x}}^1 f(x,y)dydx$$
3. It should be
$$\int_0^6 \int_0^{\frac{y}{2}} f(x,y)dxdy$$
4. It should be
$$\int_0^1 \int_{\sqrt{x}}^1 f(x,y)dydx + \int_{-1}^0 \int_{x^2}^1 f(x,y)dydx$$

5. It should be
$$\int_0^2 \int_0^{x^3} f(x,y)dydx$$
6. It should be
$$\int_{-\infty}^0 \int_0^{e^x} f(x,y)dydx + \int_0^1 \int_0^1 f(x,y)dydx$$

□

Problem 6.3. Compute the following integral:
$$\iint_D (x^2+y^2)^{1/2}dxdy$$
where D is the disk $D = \{(x,y) : x^2+y^2 \leq 9\}$.

Proof. Writing this in polar coordinates and using the change of variables formula, we have
$$\begin{aligned}\iint_D (x^2+y^2)^{1/2}dxdy &= \int_0^{2\pi}\int_0^3 (r^2)^{1/2} r dr d\theta \\ &= \int_0^{2\pi}\int_0^3 r^2 dr d\theta \\ &= 18\pi\end{aligned}$$

□

Problem 6.4 (Marsden-Tromba, 6.2, Exercise 19)**.** Compute $\iint_R (x+y)^2 e^{x-y} dxdy$, where R is the region bounded by $x+y=1, x+y=4, x-y=-1, x-y=1$.

Proof. Writing $u = x+y, v = x-y$, by the change of variables formula, we have
$$\begin{aligned}\iint_R (x+y)^2 e^{x-y} dxdy &= \int_{-1}^1 \int_1^4 u^2 e^v |\det(J)| dudv \\ &= \frac{1}{2}\left(\int_1^4 u^2 du\right)\left(\int_{-1}^1 e^v dv\right) \\ &= \frac{21}{2}\left(e - \frac{1}{e}\right)\end{aligned}$$

□

Problem 6.5. Compute $\iint_R xy^2 dA$ where R is the region bounded by $xy=1, xy=3, y=2, y=6$, using the transformation $x = \frac{v}{6u}, y = 2u$.

Proof.
$$\begin{aligned}\iint_R xy^2 dA &= \int_1^3 \int_3^9 \frac{v}{6u} 4u^2 |\det(J)| dvdu \\ &= \int_1^3 \int_3^9 \frac{2v}{9} dvdu \\ &= 2\int_3^9 \frac{2v}{9} dv \\ &= 16\end{aligned}$$

where

$$\det(J) = \det \begin{bmatrix} -\frac{v}{6u^2} & \frac{1}{6u} \\ 2 & 0 \end{bmatrix} = -\frac{1}{3u}$$

$\square$

Chapter 7

Integrals over Paths and Surfaces

Week 12 (4/6-4/10)

Topic: path integrals.

Definition 7.1 (path integral)**.** Let $c : [a, b] \to \mathbb{R}^3$ be a path of C^1 and $f : \mathbb{R}^3 \to \mathbb{R}$ is such that $f \circ c$ is continuous on $[a, b]$, The **path integral** of $f(x, y, z)$ along the path c is given by

$$\begin{aligned}\int_c f ds &= \int_a^b f(c(t))\|c'(t)\|dt \\ &= \int_a^b f(x(t), y(t), z(t))\|c'(t)\|dt\end{aligned}$$

Definition 7.2 (line integral)**.** Let F be a vector field on $\mathbb{R}^3$ that is continuous on the C^1 path $c : [a, b] \to \mathbb{R}^3$, where $c(t) = (x(t), y(t), z(t))$. We define $\int_c F \cdot ds$, the **line integral** of F along c by the following

$$\begin{aligned}\int_c F \cdot ds &= \int_a^b F(c(t)) \cdot c'(t)dt \\ &= \int_a^b \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt}\right) dt \\ &:= \int_c F_1 dx + F_2 dy + F_3 dz\end{aligned}$$

the expression $F_1 dx + F_2 dy + F_3 dz$ is called the **differential form**.

Definition 7.3 (work done along a path)**.** The force field F on a particle moving along a path $c : [a, b] \to \mathbb{R}^3$ is given by

$$\text{work done by } F = \int_a^b F(c(t)) \cdot c'(t)dt$$

Definition 7.4 (reparametrization)**.** Let $h : I \to I_1$ be a C^1 real-valued bijective function. Let $c : I_1 \to \mathbb{R}^3$ be a piecewise C^1 path. Then we call the composition

$$p = c \circ h : I \to \mathbb{R}^3$$

a **reparametrization** of c. For example, let $c : [0,1] \to \mathbb{R}^3$ be a C^1 path, then consider $h : [0,1] \to [0,1]$, where $h(t) = 1 - t$. Then the path

$$c_{\text{op}} = c \circ h(t) = c(1-t)$$

is the same path in the opposite direction.

Proposition 7.1 (reparametrization for path integrals)**.** Let c be a C^1 path and c' be any reparametrization of c, and let f be a continuous function on the image of c, then

$$\int_c f(x,y,z)ds = \int_{c'} f(x,y,z)ds$$

Proposition 7.2 (reparametrization for line integrals)**.** Let F be a vector feld continuous on the C^1 path $c : [a,b] \to \mathbb{R}^3$, and let $c' : [a',b'] \to \mathbb{R}^3$ be a reparametrization of c. If c' is orientation-preserving, then

$$\int_{c'} F \cdot ds = \int_c F \cdot ds$$

If c' is orientation-reversing, then

$$\int_{c'} F \cdot ds = -\int_c F \cdot ds$$

Proposition 7.3 (fundamental theorem of line integrals)**.** Suppose $f : \mathbb{R}^3 \to \mathbb{R}$ is of C^1 and that $c : [a,b] \to \mathbb{R}^3$ is piecewise C^1. Then

$$\int_c \nabla f \cdot ds = f(c(b)) - f(c(a))$$

Problem 7.1. Let $c : [a, b] \to \mathbb{R}^3$ be a C^1 path, find a reparametrization $\tilde{c} = c \circ h$, such that $\tilde{c} : [0, 1] \to \mathbb{R}^3$.

Problem 7.2. Suppose that $\nabla f(x, y, z) = (2x^2, y, -z)$, and that $f(0, 1, -1) = 2$. What is the value of $f(2, 1, 1)$?

Problem 7.3. Compute the path integral of $f(x, y, z) = 2x + y - z$ over the curve c, where c is the intersection of the two surfaces below:

$$y = x, \quad y^2 + z^2 = 4$$

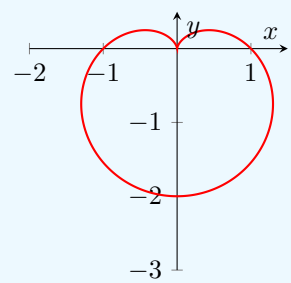
Hint: first find a parametrization of c, then use the formula of path integral.

Problem 7.4. Suppose path c has length l, and F is a vector field such that $\|F\| \leq M$. Prove that

$$\left| \int_c F \cdot ds \right| \leq Ml$$

(Hint: Cauchy-Schwarz.)

Problem 7.5. Given vector field $F(x, y) = (1, 1)$, and the following heart is given by the parametrization:

$$c(t) = ((1 - \sin t) \cos t, (1 - \sin t) \sin t) \quad \text{where } 0 \leq t \leq 2\pi$$

Group 1: what is the work done by going up the top of the heart? ($0 \leq t \leq \pi$).
Group 2: what is the work done by going up the bottom of the heart? ($\pi \leq t \leq 2\pi$).